

6	(a)	progressive waves transfer/propagate energy and stationary waves do not	B1	
		<u>amplitude</u> constant for progressive wave and varies (from max/antinode to min/zero/node) for stationary wave	B1	
		adjacent particles in phase for stationary wave and out of phase for progressive wave	(B1)	[2]
	(b)	(i)		
		wave / microwave <u>from source/S</u> reflects <u>at reflector/R</u>	B1	
		reflected and (further) incident waves overlap/meet/superpose	B1	
		waves have same <u>frequency/wavelength/period</u> and <u>speed</u> (so stationary waves formed)	B1	[3]
		(ii)		
		detector/D is moved between reflector/R and source/S (or v.v.)	B1	
		maximum, minimum/zero, (maximum... etc.) observed on meter/deflections/readings/measurements/recordings	B1	[2]
		(iii)		
		<u>determine/measure</u> the distance between adjacent minima/nodes or maxima/antinodes or across <u>specific number</u> of nodes/antinodes	B1	
		wavelength is twice distance between <u>adjacent</u> nodes/minima or maxima/antinodes (or other correct method of calculation of wavelength from measurement)	B1	[2]

Page 5	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2015	9702	22

(c) $v = f\lambda$

C1

$$f = 3.0 \times 10^8 / (2.8 \times 10^{-2}) [= 1.07 \times 10^{10} \text{ Hz}]$$

C1

11 (10.7) GHz

A1 [3]